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Studierichting: Informatica
Oefeningenreeks 3G nr. 2

$f(x) = (x-2)^3 + 1$

Nulpunten:

$f(x) = 0 \Rightarrow f(1) = (1-2)^3 + 1 = 0 \Rightarrow x = 1$

$f'(x) = 3(x-2)^2$

Extrema:

$f'(x) = 0 \Rightarrow f(2) = 3(2-2)^2 = 0 \Rightarrow x = 2$

$f''(x) = 6(x-2)$

Buigpunten:

$f''(x) = 0 \Rightarrow f''(2) = 3(2-2)^2 = 0 \Rightarrow x = 2$

Tekenonderzoek.:

x	$-\infty$	1		2	$-\infty$
$f(x)$	$-$	0	$+$	$+$	$+$
$f'(x)$	$+$	$+$	$+$	$0^{(2)}$	$+$
$f''(x)$	$-$	$-$	$-$	0	$+$
$f(x)$	$\frown$	0	$\frown$	B	$\smile$
	$\nearrow$	0	$\nearrow$	(1)	$\nearrow$

Raaklijn in punt $(1, 0)$:

$y = 3(x-1)$

References

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